

Builder Basics — The Trail Maker (Extension)

Topic: Place On Move · **Course:** Builder Basics · **Type:** Extension Activity · **Time:** about 45 minutes

✖ This is a bonus challenge, not a weekly lesson. Try it once you are comfortable placing blocks by hand and you are ready to let the agent place them for you.

1 · Predict 🧠

In the **Trail Maker** world, typing `run` starts the agent walking forward in a straight line. First it flips a switch called **place on move**.

When **place on move** is **ON**, the agent drops a block on every step — it leaves a **trail**. When it is **OFF**, the agent walks the same steps but places nothing, so a **gap** appears.

Here is the exact order the program runs:

```
place on move ON
set block to purple concrete
move forward by 4
place on move OFF
move forward by 2
place on move ON
move forward by 4
```

The switch starts ON and the agent moves forward 4. How many blocks does it drop in this first stretch?

Then the switch turns OFF and the agent moves forward 2. How many blocks does it drop while the switch is OFF?

The switch turns ON again and the agent moves forward 4. Add up every block from the whole program. How many blocks in total? Show your adding.

Describe the finished trail. Where is the gap, and how wide is it? Draw or describe the shape.

2 · Trace the Switch

Walk through the program one line at a time. For each move command, write the switch state, the steps, the blocks placed, and the running total.

Move command	Switch (ON / OFF)	Steps	Blocks this stretch	Total so far
move forward 4				
move forward 2				
move forward 4				

Which move command placed NO blocks, and why?

The agent walked forward 10 steps in total but did not place 10 blocks. Explain why the number of blocks is smaller than the number of steps.

3 · Reason About Bugs

No code to rewrite — read what goes wrong and explain it in writing.

Bug A — The switch never turns OFF. The program leaves **place on move** ON the whole time. The agent still walks the same 10 steps.

What does the trail look like now? What happened to the gap? Explain why.

Bug B — The switch never turns ON. The program forgets the first `place on move ON`, so the switch stays OFF the whole time. The agent still walks all 10 steps.

How many blocks get placed? What do you see on the ground when it finishes? Why?

4 • Design Your Own Trail

Plan a trail of your own — in words and numbers, not in a code box.

Choose your block, and decide your ON and OFF pattern. Fill in the blanks:

My block is ...

Switch ON, move forward ... (places ... blocks)

Switch OFF, move forward ... (a gap ... wide)

Switch ON, move forward ... (places ... blocks)

How many blocks does your pattern place in total? How many empty steps are in your gap?

The big idea: in Week 2 you placed each block by hand — one **place** per block. With **place** on **move** you write one **move** and the agent places a whole line. Explain WHY that is easier. If you wanted a trail 50 blocks long, which way would you rather build it, and why?

5 · Show Your Work 📷 🎥

Open the **Trail Maker** world and type **run** in the chat to watch the agent lay its trail. Then build your **own** version by changing the program — try a longer trail, a wider gap, more than one gap, or a different block.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

Write your lines here, then say them in your video.

Submit ✓

Send this worksheet + your video to teacher on KakaoTalk.